Context controls and EEG pairing in retrospective valence prediction: an exploratory FACED study

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Abstract

We evaluated whether tested electroencephalography (EEG) recipes improve individual retrospective valence prediction beyond supplied elicitation metadata on the FACED dataset. Seventy source and twenty development-validation participants formed three overlapping title-held rotations. Category and metadata controls, spectral regression and EEGNet residual prediction were examined using acquisition fallback and constrained EEG swaps. Later source diagnostics separately assessed spectral coverage on 840 held pairs and longer EEGNet learning on one 84-pair panel. All evaluations reused development data; 33 reserved participants remained sealed. Mean unfamiliar-title validation absolute error was 1.2602 for category medians, 1.2156 for spectral/context prediction and 1.2550 for EEGNet/context prediction; all corresponding descriptive gain intervals included zero. In the later source comparison, distributed spectral error was 1.4455 versus 1.3665 for matched context. On the separate neural panel, fixed 360-epoch EEGNet error was 1.4075 versus 1.2953 for context and 1.0994 for category medians. Source error continued decreasing, while real and shuffled small-set targets were fitted. Basic learning capacity therefore did not establish incremental EEG utility. These adaptive evaluations support a bounded case study, without establishing convergence, equivalence to context or absence of affective EEG information.

1 Introduction

EEG emotion benchmarks often predict assigned stimulus categories, whereas individual affect can differ within a stimulus. FACED provides elicitation categories and retrospective individual ratings [1]. Predicting individual valence requires testing whether EEG adds information beyond supplied experimental context.